

Multiple Choice (10 marks)

1. Electromagnetic radiation emitted from a nucleus is most likely to be in the form of
 - (a) gamma rays
 - (b) microwaves
 - (c) ultraviolet radiation
 - (d) visible light
2. A resistor in a circuit dissipates energy at a rate of 1 W. If the voltage across the resistor is doubled, what will be the new rate of energy dissipation?
 - (a) 0.25 W
 - (b) 0.5 W
 - (c) 1 W
 - (d) 4 W
3. A rod measures 1.00 m in its rest system. How fast must an observer move parallel to the rod to measure its length to be 0.80 m?
 - (a) 0.50 c
 - (b) 0.60 c
 - (c) 0.70 c
 - (d) 0.80 c
4. The best type of laser with which to do spectroscopy over a range of visible wavelengths is
 - (a) a dye laser
 - (b) a helium-neon laser
 - (c) an excimer laser
 - (d) a ruby laser
5. The speed of light inside of a nonmagnetic dielectric material with a dielectric constant of 4.0 is
 - (a) 1.2×10^9 m/s
 - (b) 3.0×10^8 m/s
 - (c) 1.5×10^8 m/s
 - (d) 1.0×10^8 m/s

True / False (10 marks)

Answer True or False

6. True or False: The primary source of the Sun's energy involves thermonuclear reactions between four hydrogen atoms and one helium atom.
7. True or False: The electrons in metals form a degenerate Fermi gas, leading to a higher mean kinetic energy compared to kT .
8. True or False: A helium-neon laser is the most effective laser for spectroscopy across all visible wavelengths.
9. True or False: De Broglie proposed that the linear momentum and wavelength of a free particle are related by the speed of light.
10. True or False: In the diamond structure of elemental carbon, the nearest neighbors of each C atom are located at the corners of a cube.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how the Pauli exclusion principle accounts for the energy differences between para- and ortho-states in the excited states of the helium atom.
12. Discuss the similarities between the emission spectrum of the doubly ionized lithium atom (Li^{++}) and that of a hydrogen atom, particularly focusing on the effect of atomic number on wavelength.

End of Paper